

the course of evolution. Moreover, recent phylogenomic studies indicate that turtles are nested within the diapsid clade and are more closely related to crocodilians and birds than to other reptiles (see Figure 34.25).

All turtles have a boxlike shell made of upper and lower shields that are fused to the vertebrae, clavicles (collarbones), and ribs (**Figure 34.29a**). Most of the 351 known species of turtles have a hard shell, providing excellent defense against predators. Fossil evidence shows that *Pappochelys*, a turtle that lived 240 million years ago, had a series of hard, shell-like bones along its belly. By 220 million years ago, another early turtle had a complete lower shell but an incomplete upper shell, suggesting that turtles acquired full shells in stages.

The earliest turtles could not retract their head into their shell, but mechanisms for doing so evolved independently in two separate branches of turtles. The side-necked turtles fold their neck horizontally, while the vertical-necked turtles fold their neck vertically.

Some turtles have adapted to deserts, and others live almost entirely in ponds and rivers. Still others live in the sea. Sea turtles have a reduced shell and enlarged forelimbs that function as flippers. They include the largest living turtles, the deep-diving leatherbacks, which can exceed a mass of 1,500 kg and feed on jellies. Leatherbacks and other sea turtles are endangered by being caught

in fishing nets, as well as by the residential and commercial development of the beaches where the turtles lay their eggs.

Lepidosaurs

One surviving lineage of lepidosaurs is represented by a single species, a lizard-like reptile called the tuatara (**Figure 34.29b**). Fossil evidence indicates that tuatara ancestors lived at least 220 million years ago. These organisms thrived on many continents well into the Cretaceous period and reached up to a meter in length. Today, however, tuataras are found only on 30 islands off the coast of New Zealand. When humans arrived in New Zealand 750 years ago, the rats that accompanied them devoured tuatara eggs, eventually eliminating the reptiles on the main islands. The tuataras that remain on the outlying islands are about 50 cm long and feed on insects, small lizards, and bird eggs and chicks. They can live to be over 100 years old. Their future survival depends on whether their remaining habitats are kept rat-free.

The other major living lineage of lepidosaurs consists of the lizards and snakes, or squamates, which number 10,425 species (**Figure 34.29c and d**). Many squamates are small; the Jaragua lizard, discovered recently in the Dominican Republic,

is only 16 mm long—small enough to fit comfortably on a dime. In contrast, the Komodo dragon of Indonesia is a lizard

(b) Tuatara (*Sphenodon punctatus*)



(d) Wagler's pit viper (*Tropidolaemus wagleri*)

(c) Australian thorny devil lizard (*Moloch horridus*)



(e) American alligator (*Alligator mississippiensis*)

